



Nano-zinc oxide seed treatment affects seed germination and seedling establishment under straw-induced stress in different rice genotype

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Abstract

Straw return is an important agricultural practice, however, the accumulation of allelopathic substances from straw return inhibits rice seed germination and seedling establishment, posing a critical bottleneck for its sustainable application. The application of nanomaterials in agriculture has garnered considerable attention. In particular, the use of nanoparticles for seed priming and coating represents a novel and efficient strategy to protect seedlings during their most vulnerable stage by mitigating the impacts of biotic and abiotic stresses. To investigate the effects of nano-zinc oxide (nano-ZnO) treatment and rice straw extract on rice seed germination and seedling growth, a pot experiment was conducted using two rice cultivars: Xiangyaxiangzhan and Yuxiangyouzhan. Five seed treatments (CK: No treatment, C0: 0 mg L⁻¹ nano-ZnO coating, C1: 50 mg L⁻¹ nano-ZnO coating, S0: 0 mg L⁻¹ nano-ZnO seed soaking, S1: 50 mg L⁻¹ nano-ZnO seed soaking) were designed. We evaluated seed germination, seedling growth, antioxidant responses, and photosynthetic pigment characteristics. The study revealed that nano-zinc oxide seed treatments interact with rice straw extract to differentially regulate seed germination and seedling growth in a genotype-specific manner. The germination rate of Yuxiangyouzhan was higher than that of Xiangyaxiangzhan during the germination period, but it was more inhibited by the straw extract at 7 days after sowing. Seed soaking with 50 mg·L⁻¹ nano-ZnO (S1) increased the germination rate of Xiangyaxiangzhan by up to 156.7% at 1 day after sowing. At the seedling stage, the straw extract inhibited shoot growth, while nano-ZnO coating (C1) significantly increased the root dry weight and total dry weight of Xiangyaxiangzhan. By 12 days after sowing, all seed treatments increased the root-to-shoot ratio of Xiangyaxiangzhan by 4.19% to 31.05%. Antioxidant enzyme activities (peroxidase and catalase) were variably affected by the treatments. Specifically, the S1 treatment increased superoxide dismutase activity at 5–7 days after sowing and enhanced the contents of photosynthetic pigments (chlorophyll a, chlorophyll b, carotenoids, and total chlorophyll) in both cultivars, even under high concentrations of straw extract. Correlation analysis confirmed strong positive links between early germination and subsequent biomass, and between root development and total dry weight. Based on these findings, we propose the following recommendations for practical application: In paddy fields practicing straw return, differentiated nano-ZnO seed treatment strategies should be adopted, considering the specific rice variety and residual straw concentration in the field. For fields with high straw residue or when cultivating varieties sensitive to allelopathic stress, such as 'Xiangyaxiangzhan', seed soaking with 50 mg L⁻¹ nano-ZnO is recommended to effectively ensure germination uniformity and early establishment. For varieties like 'Yuxiangyouzhan' or in fields with moderate straw levels, seed coating with 50 mg L⁻¹ nano-ZnO could be employed to promote root development and optimize the physiological status of seedlings. In practice, it is advisable to validate and optimize the nano-treatment concentration and method on a small scale, taking into account local straw return amounts, soil conditions, and the target variety. This approach would facilitate the translation of nano-priming technology into an effective agronomic measure for stabilizing seedling quality and enhancing the sustainability

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of straw return systems. This study indicates that the 50 mg L⁻¹ nano-ZnO seed treatment strategies based on variety and straw return conditions can effectively enhance seedling establishment, providing technical support for the sustainability of rice production under straw return systems.

Keywords Germination · Growth · Nano-ZnO oxide · Rice seedling · Straw extract

Introduction

Rice (*Oryza sativa* L.) serves as a staple food for more than half of the global population, making its sustainable production paramount for ensuring food security. (Lucca et al. 2006). Within this context, scientific straw management, as a key agronomic practice, plays an important role in promoting a resource-efficient and environmentally friendly rice industry (Ma et al. 2020).

Straw return, as a mainstream practice, effectively improves soil structure, enhances fertility, and bolsters ecosystem functions, representing a crucial method for nutrient cycling and utilization (Liu et al. 2023). However, this practice is not without risks. Straw decomposition in fields releases allelochemicals (e.g., phenolic acids). Their accumulation impairs soil structure and can lead to reduced crop yields (Xu et al. 2024). Continuous annual straw input can lead to reduced seedling vigor and even seedling mortality (Du et al. 2016). If not managed properly, straw incorporation can induce phytotoxic effects on following crops, as rice seedlings are highly susceptible to the allelochemicals released during decomposition (Zhang et al. 2024), severely restricting high and stable rice yields. Sensitivity to allelopathic substances varies among rice varieties, but research focused on developing effective mitigation strategies for specific varieties remains insufficient. Therefore, optimizing the proportion of straw returned and developing strategies to mitigate its allelopathic effects is crucial for sustainable rice production.

Seed priming technology, including hydro-priming and chemo-priming, is an effective pretreatment method that regulates pre-germination metabolic processes to enhance stress tolerance in seeds (Mahakham et al. 2017). Although both coating and soaking fall under the umbrella of seed treatment, their mechanisms of action are distinctly different. Soaking involves immersing seeds in a nanoparticle solution, utilizing hydration for passive water uptake; nanoparticles penetrate the seed coat and endosperm tissues along with water, directly participating in the regulation of germination metabolism, thereby achieving internal delivery and immediate activation of active components (Yang et al. 2024). In contrast, coating involves adhering nanoparticles onto the surface of dry seeds using a film-forming agent, keeping the seeds in a dormant state. Upon sowing, the film layer slowly releases nanoparticles or dissolved ions upon

contact with moisture, which are then absorbed by the roots and transported to the above-ground parts, providing a long-term, slow-release protective function (Pedrini et al. 2017).

In recent years, seed priming or coating with nanoparticles has offered an innovative approach to mitigate biotic and abiotic stresses during the seedling stage. Due to their small size effect and high surface activity, nanoparticles can penetrate the seed coat more efficiently and be absorbed by embryonic tissues, enabling targeted delivery of active components (Xue et al. 2020). Among them, nano-ZnO oxide (ZnONPs) is valued as a micronutrient fertilizer and growth regulator for its high surface activity and controlled-release properties, which enhance its efficacy (Chen et al. 2024; Sun and Hu 2025). Studies have shown that an appropriate concentration of nano-ZnO oxide (ZnO) can exert certain inhibitory effects and toxicity on rice seedlings, affecting the seed vigor index, shoot length, and root length (Yin et al. 2021). Furthermore, other studies found that both Si and nSiO₂ application increased rice dry weight, panicle number, and yield. Compared with conventional silicon fertilizer, nSiO₂ treatment significantly increased leaf chlorophyll content and photosynthesis (Elshayb et al. 2021). Li et al., (2025a, b) found that high-concentration nano-ZnO seed soaking treatment maximized the increase in germination potential and germination rate of peanut seeds under drought stress, and 50 mg L⁻¹ nano-ZnO seed soaking treatment increased POD activity and decreased MDA content in peanut leaves under drought stress. Venkidasamy et al. (2018) applied different concentrations of CuO NPs and nano-ZnO to eggplant and found that the content of non-enzymatic substances related to antioxidation increased with concentration, causing oxidative stress in plants. However, existing studies largely focus on the responses of specific crops under single stress conditions. There is a lack of systematic comparison and elucidation regarding the effectiveness of nano-ZnO seed treatments (such as coating and soaking) in mitigating straw allelopathic stress—specifically, whether their effects are variety-specific, the extent to which their action is attributable to nano-specific properties rather than mere zinc nutrition, and potential differences in effects compared to other metal oxide nanomaterials.

Based on the aforementioned knowledge gap, this study aims to investigate the effectiveness of nano-ZnO seed treatments (coating and soaking) in alleviating the allelopathic inhibitory effects of rice straw extract and their

genotype-dependent nature. This will be achieved by analyzing the effects of different treatments on the germination characteristics, seedling growth, and physiological-biochemical traits of two rice varieties. The research is expected to provide a theoretical basis and practical strategy for establishing a precise seedling regulation technology that synergizes straw return and nano-priming, thereby offering technical support for overcoming the application bottleneck of straw return and promoting the sustainability of rice production systems.

Materials and methods

Experimental descriptions

In this experiment, acid-washed and sterilized quartz sand (autoclaved at high temperature) was used as the culture medium. The two widely cultivated rice varieties used in the study, Yuxiangyouzhan and Xiangyaxiangzhan, were provided by the Rice Research Laboratory of South China Agricultural University (SCAU). These two aromatic rice varieties were selected due to their significant differences in seed germination ability and seedling growth performance under stress conditions. Seeds were germinated in commercial plastic pots (16 cm length $\times$ 12 cm width $\times$ 11 cm height). Zinc oxide nanoparticles (Nano-ZnO, purity 99.9%, particle size 30 ± 10 nm) were purchased from Aladdin Biochemical Technology Co., Ltd., Shanghai, China. (Li et al. 2021). The characteristics of the Nano-ZnO used have been previously reported: the average particle size was determined to be 36 nm by scanning electron microscopy (SEM) and 28 nm by transmission electron microscopy (TEM), which are consistent with the manufacturer's specifications (~ 30 nm) (Huang et al. 2020).

Experimental design

This experiment included two rice varieties (Xiangyaxiangzhan and Yuxiangyouzhan), three concentrations of rice straw extract (0 g L⁻¹, 10 g L⁻¹, and 20 g L⁻¹) (denoted as L, M, and H, respectively), and five seed treatments: CK (no treatment), C0 (0 mg L⁻¹ nano-ZnO coating, coating control), C1 (50 mg L⁻¹ nano-ZnO coating), S0 (0 mg L⁻¹ nano-ZnO seed soaking, soaking control), and S1 (50 mg L⁻¹ nano-ZnO seed soaking). The term seed soaking is used consistently for S0 and S1 treatments. To ensure sufficient sampling, six replicate pots were set up for each treatment. The concentrations of the straw extract and Nano-ZnO were determined based on previous studies (Liu et al. 2016; Zhao et al. 2021).

The straw extract was prepared according to the method of Yu et al. (2013). Mature, dry, and non-decomposed rice straw was selected and cut into 2–5 cm segments using scissors. Subsequently, 0 g, 10 g, and 20 g of straw were weighed and added to 1 L of Kimura B nutrient solution, respectively, and allowed to soak for 3 days, after which all solid straw residues were removed. The Kimura B medium was prepared following the method of Qing et al. (2022). For the nano-ZnO seed soaking treatment, seeds of Xiangyaxiangzhan and Yuxiangyouzhan were immersed in a 2% hydrogen peroxide solution for 10 min, then rinsed four times with pure water and dried. Different masses (0 mg and 50 mg) of ZnO nanoparticles (NPs) were weighed and added to separate beakers. Subsequently, 1 L of water was added to each beaker, and the beakers were placed in an ultrasonic oscillator (25 °C, 800 W, 40 kHz) for 1 h. An appropriate amount of Tween 80 was added to ensure the solution was fully dissolved. The nano-ZnO solution (pure water for the control group) was added to the surface-dried seeds, followed by the addition of a coating agent (containing 1 g chitosan, 1 g polyvinyl alcohol, and 0.1 g safranin T). The seeds were uniformly coated by manually shaking them in the container for 2 min. After 4 days, the seeds were spread on filter paper and placed in an incubator (40 °C) for 3 h, during which time the seeds were intermittently stirred to prevent adhesion. The following day, 250 treated seeds were sown per pot in pots filled with quartz sand. The coated seeds were placed in pots with dimensions of 16 cm in length, 12 cm in width, and 11 cm in height. A piece of germination paper (11 cm $\times$ 11 cm) was placed in each pot, and 10 mL of straw extract at different concentrations was added. The pots were placed in a growth chamber set to a relative humidity of 70%, with a photoperiod of 14 h light/28 °C and 10 h dark/25 °C, and a daytime photosynthetically active radiation intensity of 400 $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$ (Li et al. 2021). Straw extract was added daily to ensure the seeds remained moist.

Sampling and measurements

Determination of germination rate

Germination was defined as the point when the radicle length exceeded 2 mm. Seed germination was observed and recorded at the 1, 3, 5, and 7 days after sowing (d). The germination rate was calculated using the following formula:

Germination rate (%)

$$= (\text{number of germinated seeds} / \text{total number of test seeds}) \times 100\%.$$

Determination of plant height, root length and biomass

At 7 days after sowing, the fresh weight of the whole seedling (Due to the minimal biomass of seedlings during the early growth stage, seeds were included in the measurement to facilitate the observation of treatment effects.) was measured. For each treatment, three pots were randomly harvested, with five plants sampled per pot. Each plant was then immediately dissected into three components: seed, root, and shoot. The root and shoot portions were oven-dried to determine their dry weights. The total plant dry weight was calculated as the sum of the root dry weight and the shoot dry weight. All biomass data were recorded in milligrams (mg). Additionally, plant height and root length were measured at 7 days after sowing using a ruler, and both parameters were recorded in centimeters (cm).

At 12 days after sowing, uniform seedlings were harvested. Three pots were randomly sampled per treatment, with five plants measured per pot. The seedlings were immediately separated into roots and shoots. The fresh and dry weights of the roots and shoots were recorded, and plant height was measured using a ruler. The following parameters were then calculated: total plant fresh weight, total plant dry weight, root to shoot ratio (based on dry weight), and dry weight per unit plant height.

Determination of antioxidant attributes and photosynthetic pigments

The experimental design aimed to compare the variations among treatments within the same variety at a single time point. Whole seedlings sampled at 5 d were used to measure SOD, POD, and CAT activity, while those sampled at 7 d were used for SOD and POD activity and MDA content. At 12 d, the shoot portion of seedlings was collected for determination of antioxidant attributes (SOD, POD, and CAT activity, and MDA content) and photosynthetic pigments (chlorophyll a, chlorophyll b, carotenoids, and total chlorophyll content). All samples were immediately frozen in liquid nitrogen and stored at -80°C . Frozen tissue (0.15 g) was homogenized in liquid nitrogen, mixed with 1.5 mL of 0.05 M phosphate buffer (PBS, pH 7.8), and subjected to ultrasonic extraction in an ice bath for 10 min. The homogenate was centrifuged at 8000 r min for 15 min at 4°C , and the supernatant was collected as the enzyme extract. The antioxidant attributes and photosynthetic pigments were measured according the method of Li et al. (2021).

Superoxide dismutase (SOD) activity was determined by the NBT photoreduction method. One unit (U) of SOD activity was defined as the amount of enzyme causing 50% inhibition of NBT reduction. Peroxidase (POD) activity was measured using guaiacol as the substrate. The reaction

system was monitored at 470 nm every 30 s for 2.5 min. One unit of POD activity was defined as a ΔA_{470} of 0.01 per minute. Catalase (CAT) activity was determined by the ammonium molybdate method. Absorbance was measured at 405 nm after 10 min, with one unit defined as the amount of enzyme decomposing 1 $\mu\text{mol H}_2\text{O}_2$ per minute. Malondialdehyde (MDA) content was assessed using the thiobarbituric acid (TBA) method. A mixture of 0.2 mL enzyme extract and 0.4 mL 0.5% TBA was heated in a boiling water bath for 30 min, then cooled and centrifuged. Absorbance was measured at 532, 600, and 450 nm for calculation. All antioxidant enzyme activities (SOD, POD, and CAT) are expressed as units per gram fresh weight (U g^{-1}). MDA content is expressed as $\mu\text{mol g}^{-1}$.

For photosynthetic pigments, fresh shoot tissue (0.1 g) was extracted with 6 ml of 95% ethanol and stored in darkness at 4°C for 24 h. After centrifugation, the absorbance of the supernatant was recorded at 665, 649, and 470 nm. Pigment concentrations (chlorophyll a, b, total chlorophyll, and carotenoids) were then calculated.

Data analysis

Data analysis was performed using Microsoft Excel 2019, Statistics 8 software, and Origin. All data are presented as means. Analysis of variance (ANOVA) was conducted, and multiple comparisons were performed using the Least Significant Difference (LSD) test at a significance level of $p < 0.05$. The Correlation analysis and associated plots were generated using Origin 2025, while box plots were created with R software.

Results

Germination rate

Variety (V), straw extract (E), seed treatment (T), and their interactions ($V \times E$, $V \times T$, $E \times T$, $V \times E \times T$) had significant effects on the germination rate from 1 to 7 d, except for the $V \times E$ interaction at 3 d (Table 1). At each observation time point, the mean germination rate of Yuxiangyouzhan was higher than that of Xiangyaxiangzhan, with increases of 75.9%, 12.40%, 7.49%, and 6.37% at 1, 3, 5, and 7 days, respectively. The germination rate was reduced because of straw extract treatment, specifically in the early germination period. Nano-ZnO oxide seed treatments significantly increased the average germination rate of Xiangyaxiangzhan throughout the germination period. In contrast, for Yuxiangyouzhan, only the S0 and S1 treatments enhanced germination at 3 and 5 days. The efficacy of the seed treatment was contingent upon the concentration of straw extract and was also

Table 1 Analysis of variance (ANOVA) of the investigated parameters

Parameter	Variety (V)	Straw extract (E)	Seed treatment (T)	V×E	V×T	E×T	V×E×T
Germination at 1 day	**	**	**	**	**	**	**
Germination at 3 days	**	**	**	ns	**	**	**
Germination at 5 days	**	**	**	**	**	**	**
Germination at 7 days	**	**	**	**	**	**	**
Plant-seed fresh weight at 7 days	**	**	**	*	*	**	ns
Shoot dry weight at 7 days	*	ns	**	*	**	**	ns
Root dry weight at 7 days	**	**	**	**	**	**	**
Total plant dry weight at 7 days	**	**	**	**	**	**	**
Plant height at 7 days	ns	ns	**	**	*	ns	ns
Root length at 7 days	*	**	**	**	**	**	**
Shoot fresh weight at 12 days	**	ns	*	ns	*	*	ns
Root fresh weight at 12 days	**	**	**	ns	**	**	**
Total plant fresh weight at 12 days	**	**	**	ns	**	**	**
Shoot dry weight at 12 days	**	**	*	**	**	**	**
Root dry weight at 12 days	*	**	**	**	**	**	**
Total plant dry weight at 12 days	**	**	*	**	**	**	**
Plant height at 12 days	*	**	*	**	**	**	**
Root to shoot ratio at 12 days	*	**	**	**	**	**	**
Dry weight per plant height at 12 days	**	ns	**	**	*	**	**
SOD activity at 5 days	*	**	**	ns	**	**	**
POD activity at 5 days	*	**	**	ns	**	**	*
CAT activity at 5 days	**	**	**	*	**	**	**
SOD activity at 7 days	ns	**	**	**	**	**	**
POD activity at 7 days	*	*	**	**	**	**	**
MDA content at 7 days	**	**	**	**	**	**	**
SOD activity at 12 days	ns	ns	**	ns	ns	ns	ns
POD activity at 12 days	**	*	**	**	**	**	**
CAT activity at 12 days	*	**	**	*	**	**	**
MDA content at 12 days	ns	**	**	**	**	**	**
Chlorophyll a content at 12 days	**	**	**	**	**	**	**
Chlorophyll b content at 12 days	**	ns	**	**	ns	**	**
Carotenoids content at 12 days	*	**	**	**	**	**	**
Total chlorophyll content at 12 days	**	**	**	**	ns	**	**

*Significant at $p < 0.05$; **, significant at $p < 0.01$; ns, nonsignificant at the $p > 0.05$ level

variety-specific. On average, nano-ZnO seed soaking treatment (S1) performed better in Xiangyaxiangzhan, increasing the germination rate by 156.7%, 6.81%, 5.64%, and 6.13% at 1 d, 3 d, 5 d, and 7 d compared with the CK, respectively (Fig. 1). Compared to CK, seed treatments enhanced the germination rate of Xiangyaxiangzhan at 1 d across all straw extract levels (L, M, H). In contrast, for Yuxiangyouzhan, only treatments C0, C1, and S0 improved germination under the high extract level (H), while other treatments either reduced or did not significantly change the germination rate at 1 d (Fig. 1A). At the moderate straw extract level (M), seed treatments enhanced the germination rate of Xiangyaxiangzhan at 3d. At the high level (H), treatments S0 and S1 improved germination rate at 3 d

in both Xiangyaxiangzhan and Yuxiangyouzhan, respectively (Fig. 1B). At 5 d, the germination response varied with straw extract concentration: under the L level, treatment S1 significantly increased the germination rate of Xiangyaxiangzhan; under the M level, all seed treatments were effective for Xiangyaxiangzhan; and under the H level, treatments C1, S0 and S1 enhanced germination in Xiangyaxiangzhan, whereas for Yuxiangyouzhan, the germination rate decreased or showed no significant change under low and moderate levels, and the S0 and S1 treatments increased the germination rate under the high level (Fig. 1C). At 7 d, all seed treatments enhanced the germination rate of Xiangyaxiangzhan under the M straw

Fig. 1 Effects of seed treatment and straw extract on rice germination rate. Straw extract concentrations: L (0 g L^{-1}), M (10 g L^{-1}), H (20 g L^{-1}). Seed treatments: CK (no treatment), C0 (0 mg L^{-1} coating), C1 (50 mg L^{-1} coating), S0 (0 mg L^{-1} seed soaking), S1 (50 mg L^{-1} seed soaking). Different lowercase letters above bars indicate significant differences among treatments within the same variety (LSD test, $p < 0.05$)

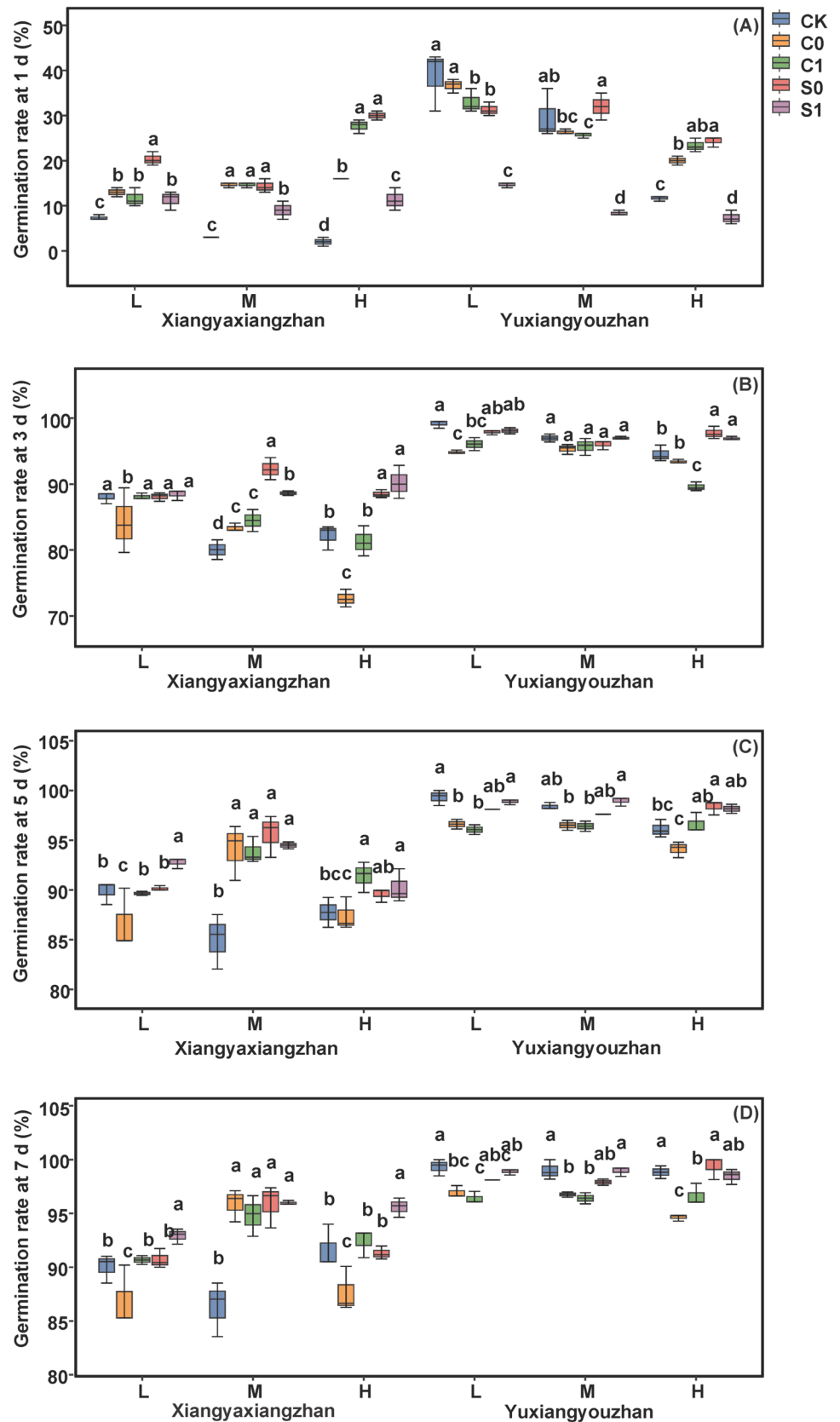
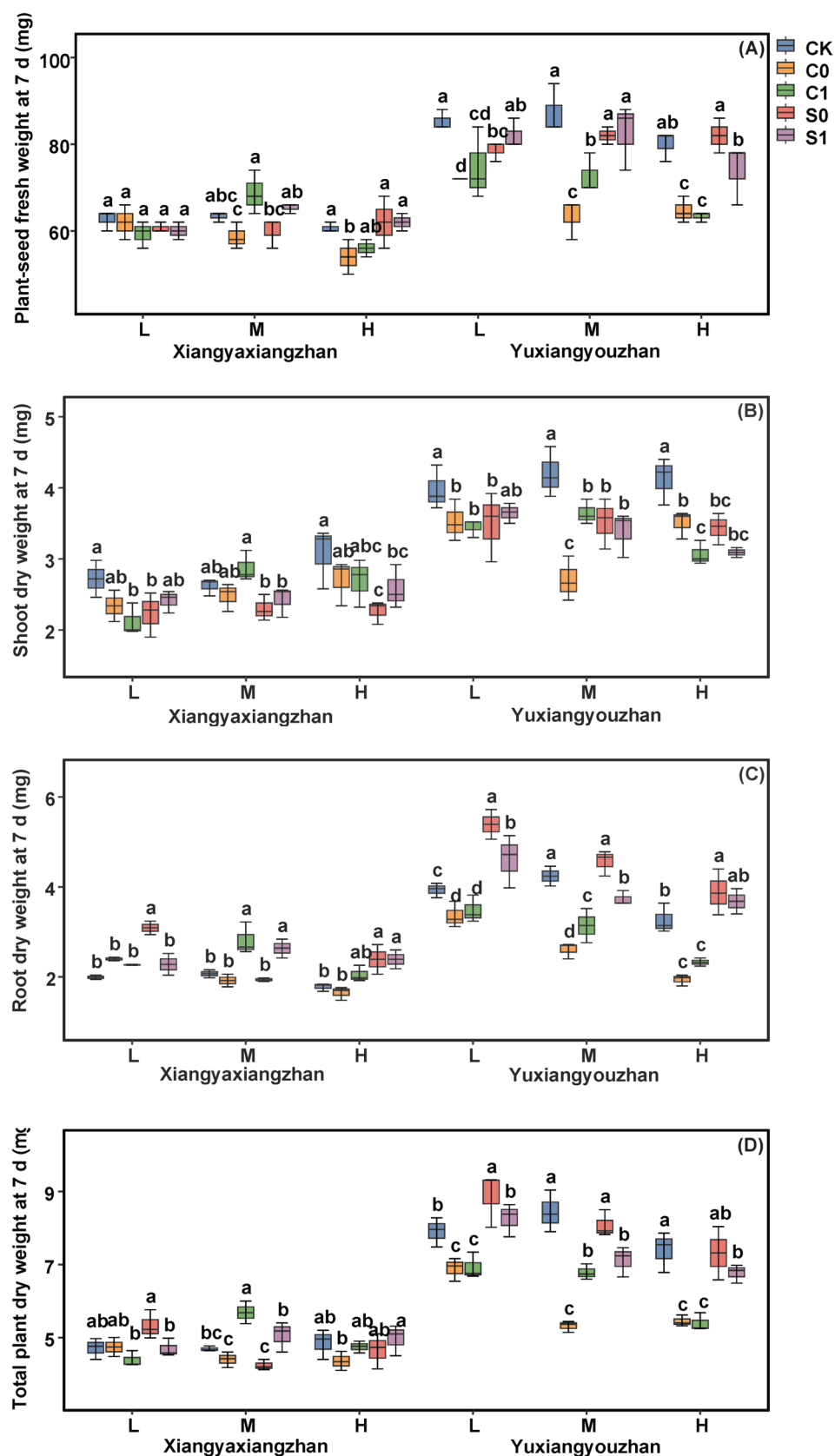


Fig. 2 Effects of seed treatment and straw extract on biomass in rice seedlings at 7 days after treatment (7 days). Straw extract concentrations: L (0 g L^{-1}), M (10 g L^{-1}), H (20 g L^{-1}). Seed treatments: CK (no treatment), C0 (0 mg L^{-1} coating), C1 (50 mg L^{-1} coating), S0 (0 mg L^{-1} seed soaking), S1 (50 mg L^{-1} seed soaking). Different lowercase letters above bars indicate significant differences among treatments within the same variety (LSD test, $p < 0.05$)



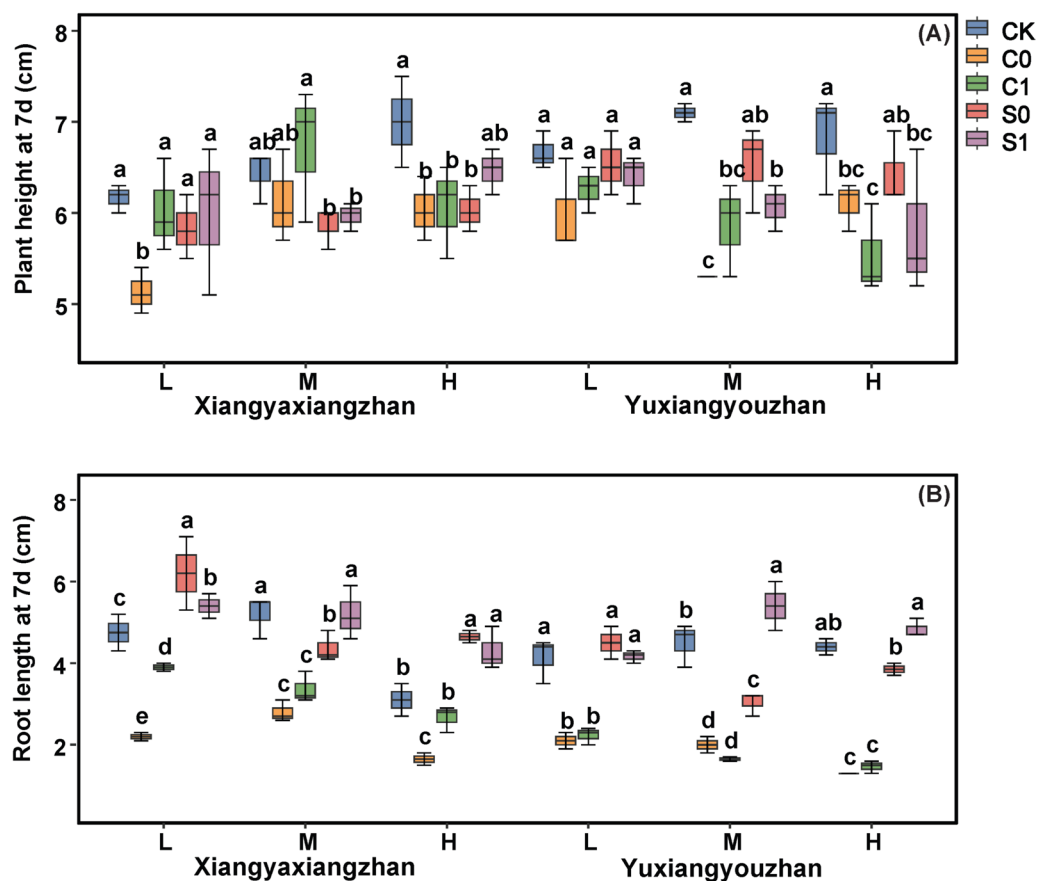


Fig. 3 Effects of seed treatment and straw extract on plant height, and root length of rice seedlings at 7 days after treatment (7 days). Straw extract concentrations: L (0 g L⁻¹), M (10 g L⁻¹), H (20 g L⁻¹). Seed treatments: CK (no treatment), C0 (0 mg L⁻¹ coating), C1 (50 mg L⁻¹

coating), S0 (0 mg L⁻¹ seed soaking), S1 (50 mg L⁻¹ seed soaking). Different lowercase letters above bars indicate significant differences among treatments within the same variety (LSD test, $p < 0.05$)

extract level. Under the H level, treatment S1 also significantly increased the germination rate, respectively (Fig. 1D).

Growth parameters

V and E had a significant or extremely significant effect on almost all seedling growth parameters, and T as well as the V × T interaction also exerted a significant regulatory effect on seedling growth parameters (Table 1). The growth responses of rice to straw extract and nano-ZnO oxide seed treatment exhibited distinct temporal specificity (7 d and 12 d) and genotypic specificity, with a stable quantitative difference in biomass accumulation between the two cultivars (Figs. 2, 3). For growth parameter at 7d, although Yuxiangyouzhan had higher biomass than Xiangyaxiangzhan, its growth was more significantly inhibited by straw extract treatment. Compared with the CK, seed treatments under different concentrations of straw extract all reduced the plant-seed fresh weight, shoot dry weight and plant height, and the efficacy of nano-ZnO oxide seed treatments was

affected by straw extract. The nano-ZnO coating treatment (C1) increased the average root dry weight and total dry weight of Xiangyaxiangzhan. Seed soaking treatments (S0 and S1) increased the average root dry weight of both cultivars and enhanced the root length of Xiangyaxiangzhan and Yuxiangyouzhan. Specifically, the S1 treatment additionally increased the mean total plant dry weight in Xiangyaxiangzhan and the root length in Yuxiangyouzhan. In Xiangyaxiangzhan, compared with the CK, root dry weight was increased by the S0 and S1 treatments under the low level, by the C1 and S1 treatments under the moderate level, and by the S0 and S1 treatments under the high level; total dry weight was elevated by the S0 treatment under the low level and by the C1 and S1 treatments under the moderate level; root length was increased by the S0 and S1 treatments under both low and high levels, but decreased or showed no significant change under the moderate level. In Yuxiangyouzhan, root dry weight was improved by the S0 and S1 treatments under both low and high levels; root length was increased by the S1 treatment under moderate and high levels (Fig. 3).

As seedlings developed to 12 days after sowing, the growth inhibition caused by straw extract was alleviated in both cultivars (except for root dry weight), and Yuxiangyouzhan outperformed Xiangyaxiangzhan in most growth parameters. Notably, straw extract treatment increased the plant height of both cultivars, and also elevated the average shoot dry weight and dry weight per unit plant height

of Xiangyaxiangzhan. Compared to CK, the C1 treatment increased shoot and total fresh weight in Xiangyaxiangzhan. The S1 treatment was more comprehensive, enhancing total fresh weight, root dry weight, and dry weight per plant height in Xiangyaxiangzhan, while also increasing the root to shoot ratio in Yuxiangyouzhan. Notably, in Xiangyaxiangzhan, all seed treatments consistently elevated the

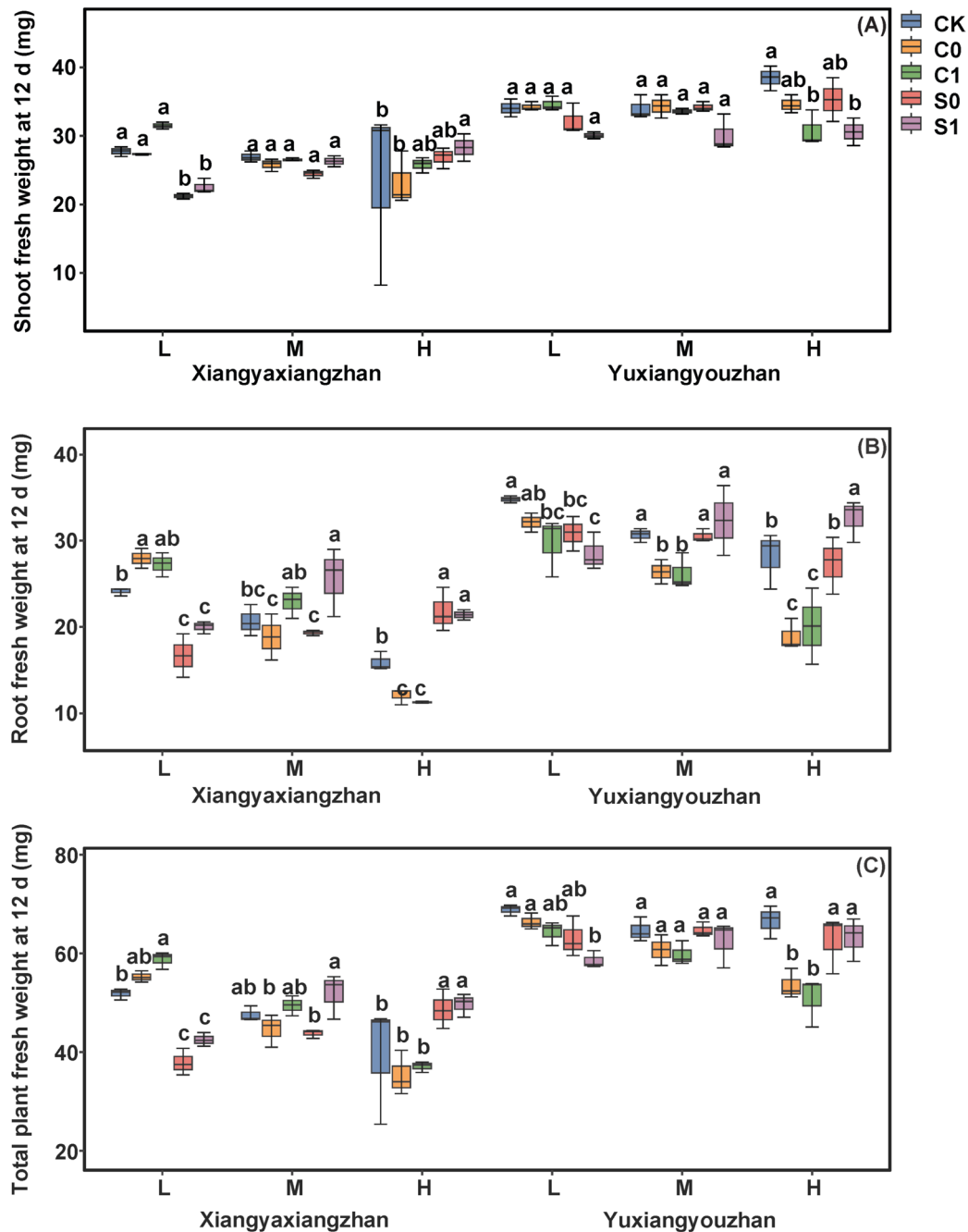


Fig. 4 Effects of seed treatment and straw extract on shoot fresh weight, root fresh weight, and total fresh weight at 12 days after treatment (12 days). Straw extract concentrations: L (0 g L⁻¹), M (10 g L⁻¹), H (20 g L⁻¹). Seed treatments: CK (no treatment), C0 (0 mg

L⁻¹ coating), C1 (50 mg L⁻¹ coating), S0 (0 mg L⁻¹ seed soaking), S1 (50 mg L⁻¹ seed soaking). Different lowercase letters above bars indicate significant differences among treatments within the same variety (LSD test, $p < 0.05$)

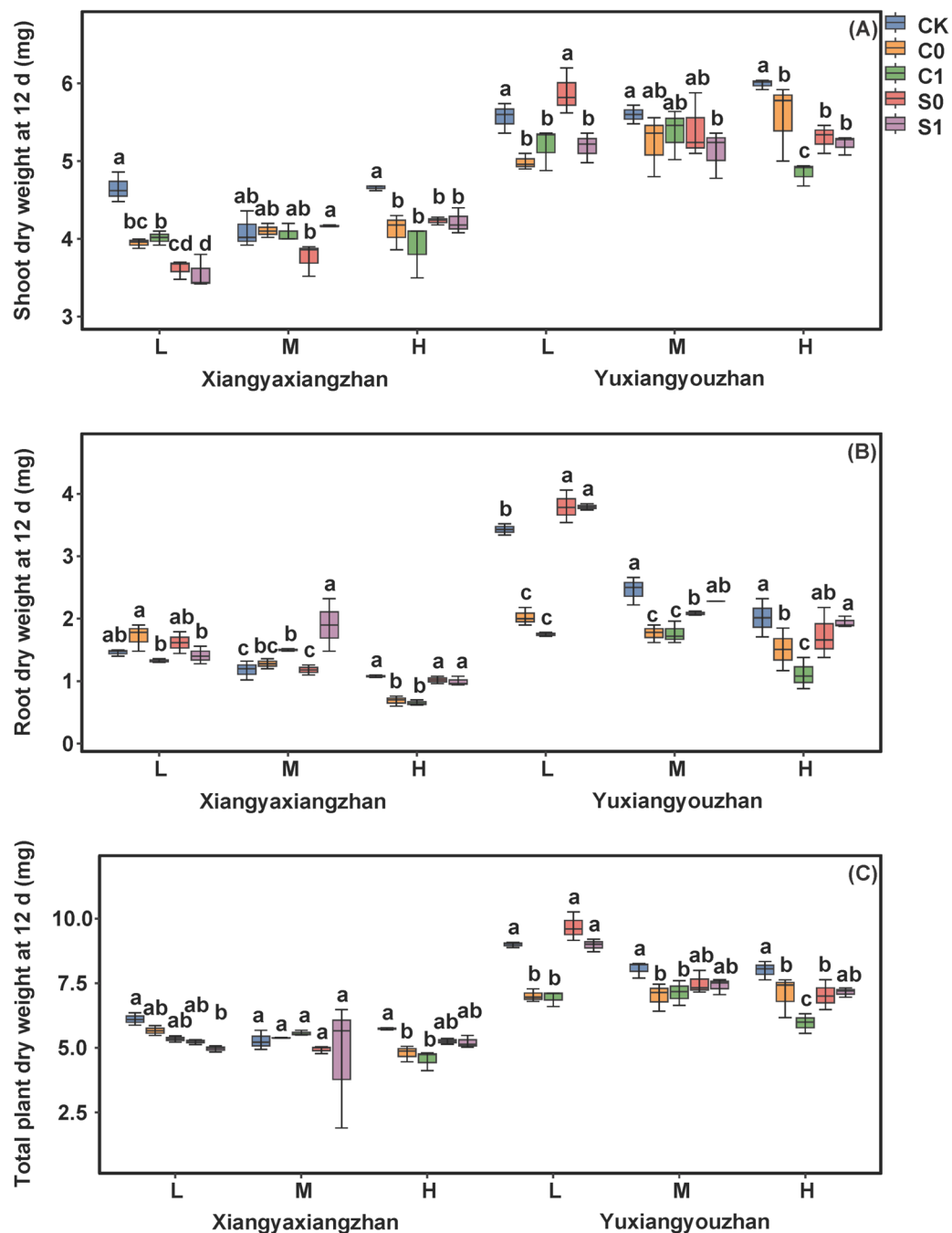


Fig. 5 Effects of seed treatment and straw extract on shoot dry weight, root dry weight, and total dry weight at 12 days after treatment (12 days). Straw extract concentrations: L (0 g L⁻¹), M (10 g L⁻¹), H (20 g L⁻¹). Seed treatments: CK (no treatment), C0 (0 mg

L⁻¹ coating), C1 (50 mg L⁻¹ coating), S0 (0 mg L⁻¹ seed soaking), S1 (50 mg L⁻¹ seed soaking). Different lowercase letters above bars indicate significant differences among treatments within the same variety (LSD test, $p < 0.05$)

root-to-shoot ratio, with increases ranging from 4.19 to 31.05%. The effects of seed treatments on seedling growth exhibited clear cultivar specificity. In Xiangyaxiangzhan, total fresh weight was enhanced by the C0 and C1 treatments

under the low straw extract level, and by the S0 and S1 treatments under the high level. The root to shoot ratio was enhanced by C0, S0, and S1 at the L level, and by C1 and S1 at the M level. Additionally, dry weight per unit plant height

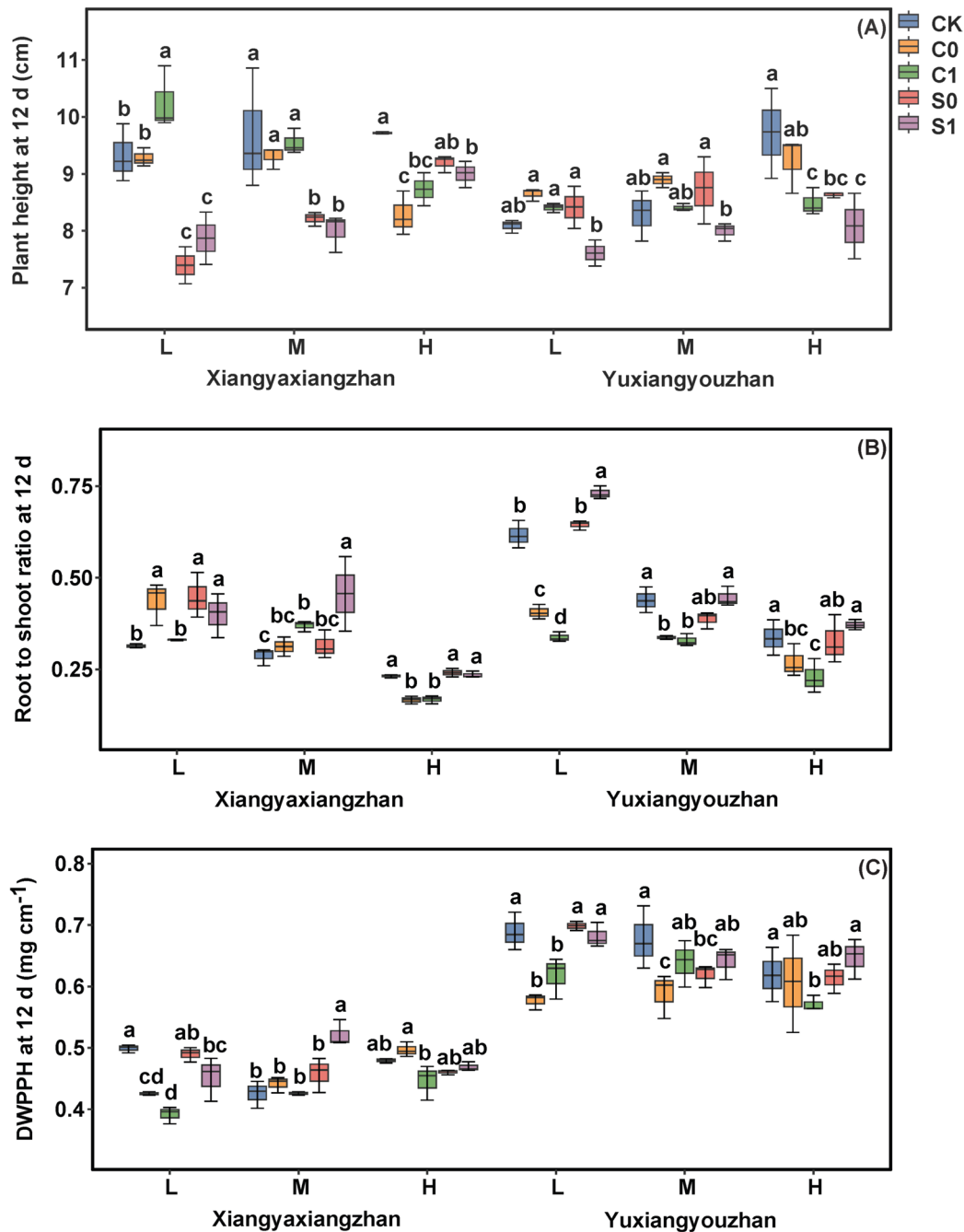


Fig. 6 Effects of seed treatment and straw extract on plant height, root to shoot ratio, and dry weight per plant height at 12 days after treatment (12 days). Straw extract concentrations: L (0 g L⁻¹), M (10 g L⁻¹), H (20 g L⁻¹). Seed treatments: CK (no treatment), C0

(0 mg L⁻¹ coating), C1 (50 mg L⁻¹ coating), S0 (0 mg L⁻¹ seed soaking), S1 (50 mg L⁻¹ seed soaking). Different lowercase letters above bars indicate significant differences among treatments within the same variety (LSD test, $p < 0.05$)

increased with S1 at the L level. In Yuxiangyouzhan, total fresh weight was reduced by all seed treatments under all straw extract concentrations. root dry weight was improved by S0 and S1 at the L level, and the root-to-shoot ratio was elevated by S1, also at the L level (Figs. 2, 3).

Antioxidant response

V, E, T, and their interactions had a significant or extremely significant effect on most antioxidant response parameters (Table 1). Antioxidant enzyme activities exhibited temporal and tissue specificity. Obvious changes in enzyme

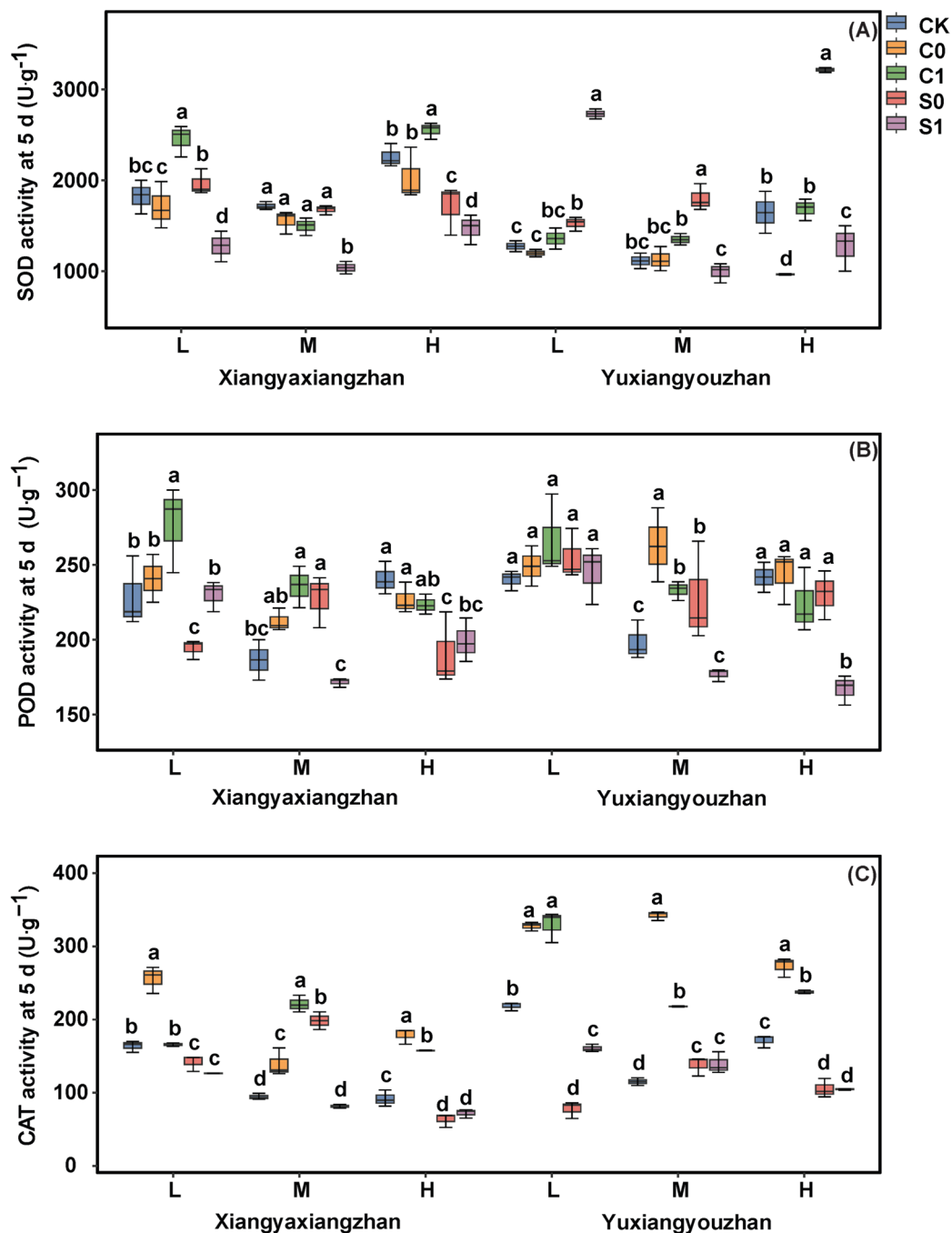


Fig. 7 Effects of seed treatment and straw extract on the antioxidant response of rice seedlings at 5 days after treatment (5 days). Superoxide dismutase (SOD) activity (A), peroxidase (POD) activity (B), and catalase (CAT) activity (C) in the whole plant at 5 days after treatment (5 days). Straw extract concentrations: L (0 g L^{-1}), M (10 g L^{-1}),

H (20 g L^{-1}). Seed treatments: CK (no treatment), C0 (0 mg L^{-1} coating), C1 (50 mg L^{-1} coating), S0 (0 mg L^{-1} seed soaking), S1 (50 mg L^{-1} seed soaking). Different lowercase letters above bars indicate significant differences among treatments within the same variety (LSD test, $p < 0.05$)

activity levels and inter-cultivar differences were observed when sampling the whole plant at 5 and 7 d and sampling only the shoot tissue at 12 d (Figs. 4, 5, 6, 7, 8, 9). At 5 d, the activities of POD and CAT in the whole plant were higher in Yuxiangyouzhan than in Xiangyaxiangzhan. By contrast, at 12 d, the activities of these enzymes in the

shoot tissue were lower in Yuxiangyouzhan than in Xiangyaxiangzhan. The antioxidant response varied significantly with both rice variety and straw extract level. Specifically, at 5 d, for Xiangyaxiangzhan, SOD activity was increased by the C1 treatment under low and high levels, while the S1 treatment decreased SOD activity under all extract levels;

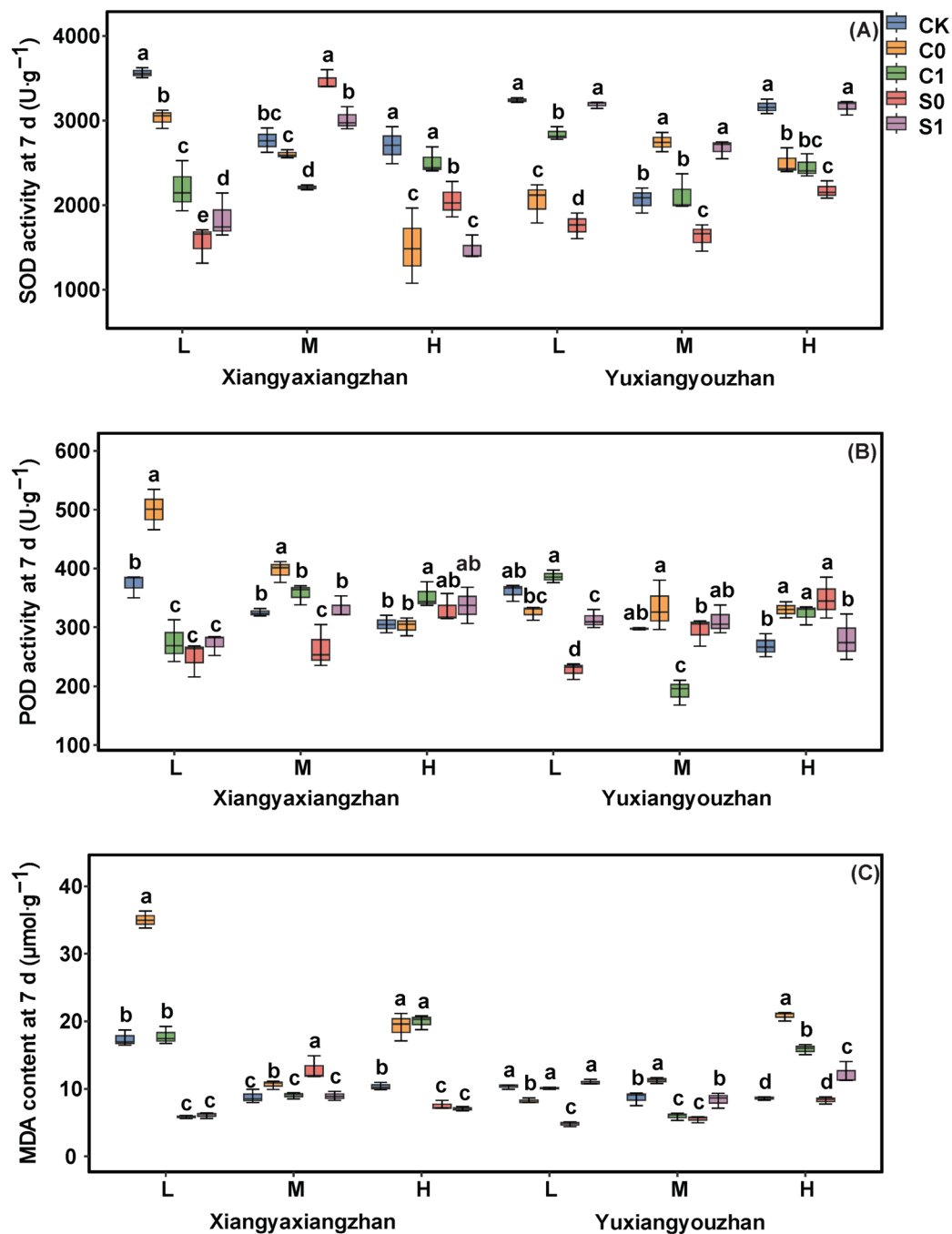


Fig. 8 Effects of seed treatment and straw extract on the antioxidant response of rice seedlings at 7 days after treatment (7 days). Super-oxide dismutase (SOD) activity (A), peroxidase (POD) activity (B), and malondialdehyde (MDA) content (C) in the whole plant at 7 days after treatment (7 days). Straw extract concentrations: L (0 g L⁻¹),

M (10 g L⁻¹), H (20 g L⁻¹). Seed treatments: CK (no treatment), C0 (0 mg L⁻¹ coating), C1 (50 mg L⁻¹ coating), S0 (0 mg L⁻¹ seed soaking), S1 (50 mg L⁻¹ seed soaking). Different lowercase letters above bars indicate significant differences among treatments within the same variety (LSD test, $p < 0.05$)

meanwhile, the C1 treatment elevated the average activities of POD and CAT. For Yuxiangyouzhan, SOD activity was improved by the S0 and S1 treatments under the low level and by the S0 treatment under moderate and high levels; the C0 and C1 treatments increased the average activities of POD and CAT (Figs. 4, 5, 6). At 7 d, the average SOD

activity of both cultivars decreased with all seed treatments under low and high straw extract levels. The C0 treatment increased the POD activity and MDA content of Xiangyaxiangzhan under low and moderate levels, and the MDA content of Yuxiangyouzhan under moderate and high levels (Figs. 7, 8). At 12 d, the S1 treatment improved the POD

Fig. 9 Effects of seed treatment and straw extract on the antioxidant response of rice seedlings at 12 days after treatment (12 days). Superoxide dismutase (SOD) activity (**A**), peroxidase (POD) activity (**B**), catalase (CAT) activity (**C**), and malondialdehyde (MDA) content (**D**) at 12 days after treatment (12 days). Straw extract concentrations: L (0 g L^{-1}), M (10 g L^{-1}), H (20 g L^{-1}). Seed treatments: CK (no treatment), C0 (0 mg L^{-1} coating), C1 (50 mg L^{-1} coating), S0 (0 mg L^{-1} seed soaking), S1 (50 mg L^{-1} seed soaking). Different lowercase letters above bars indicate significant differences among treatments within the same variety (LSD test, $p < 0.05$)

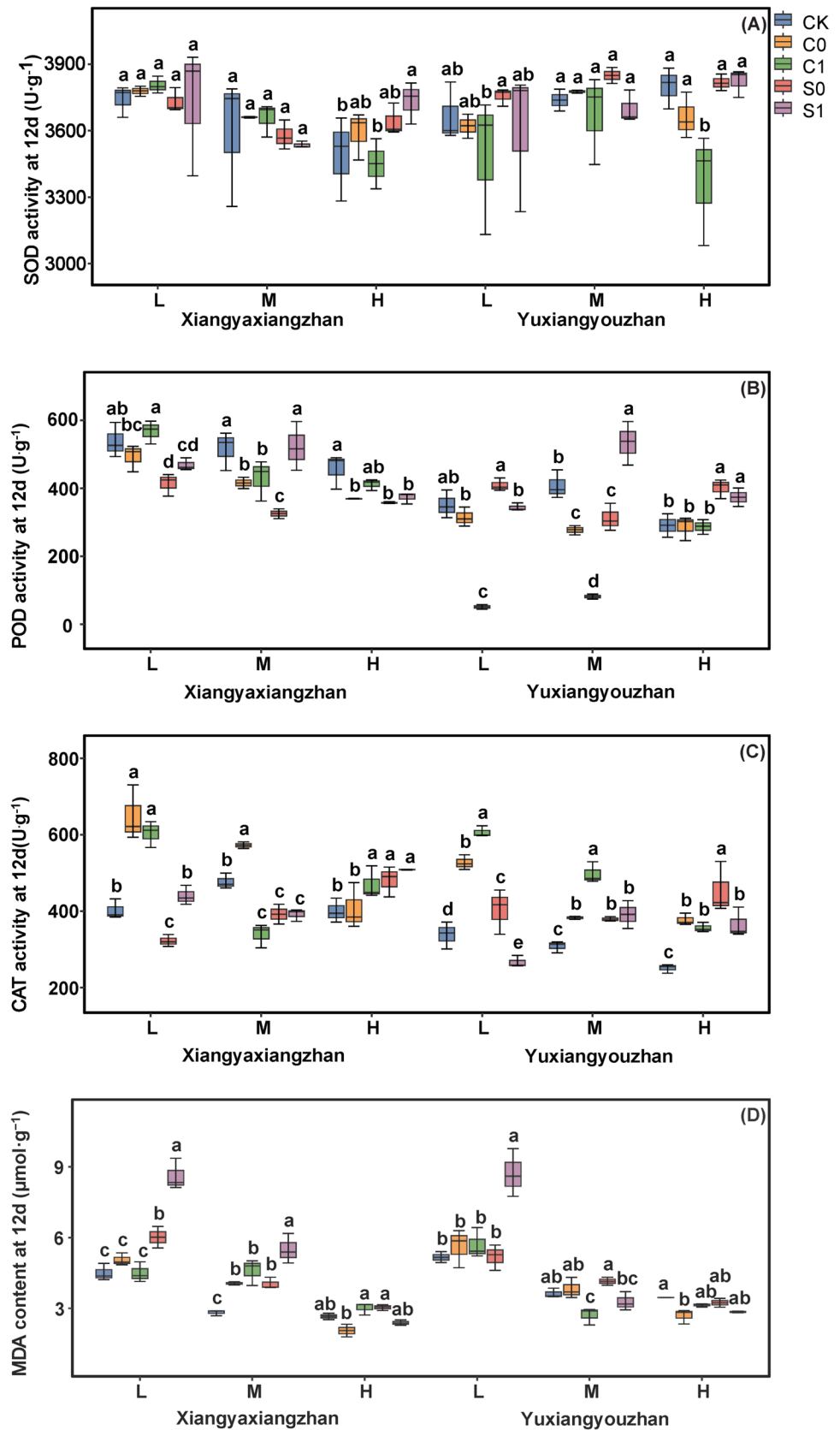
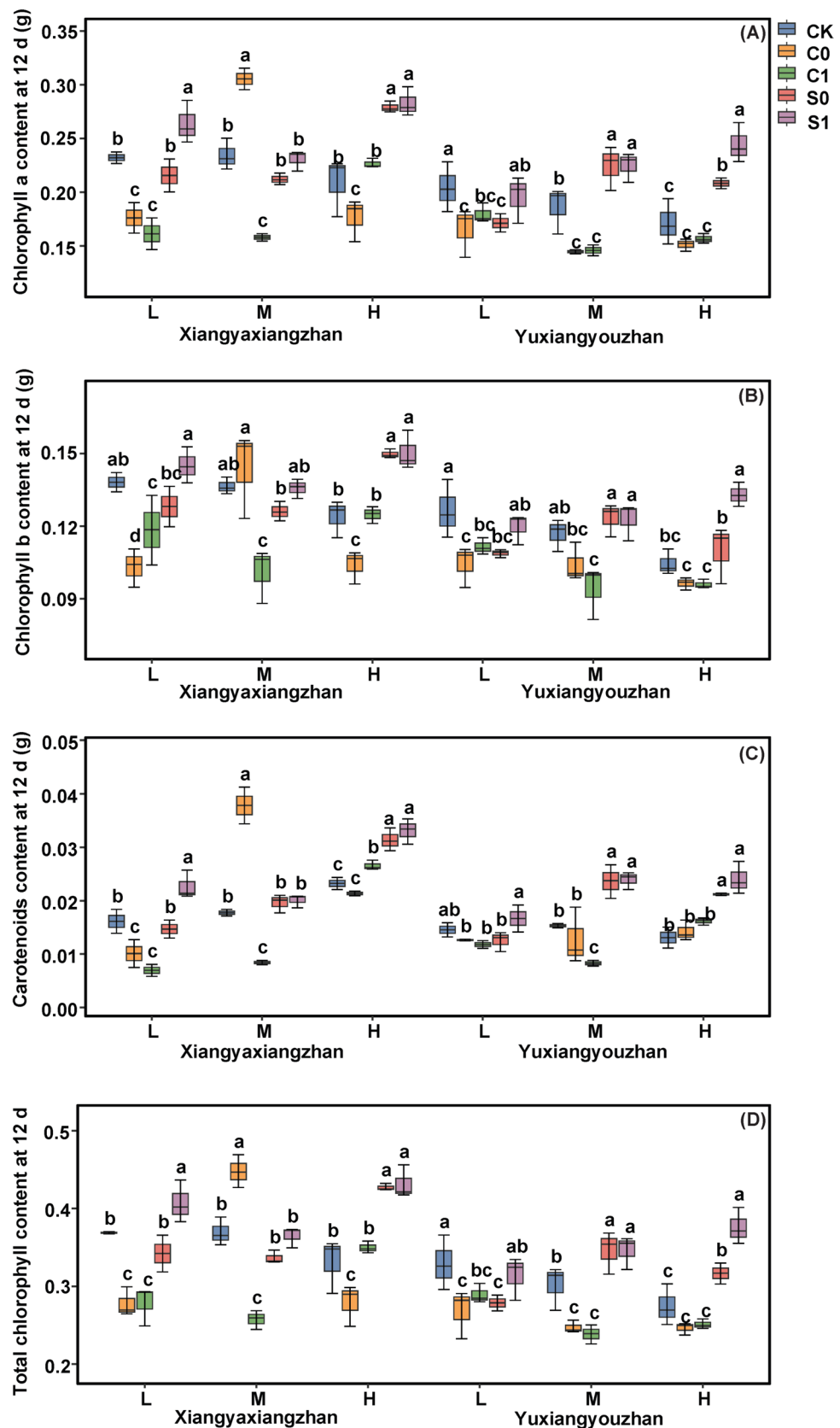


Fig. 10 Effects of seed treatment and straw extract on the photosynthetic pigments of rice seedlings at 12 days after treatment (12 days). Straw extract concentrations: L (0 g L^{-1}), M (10 g L^{-1}), H (20 g L^{-1}). Seed treatments: CK (no treatment), C0 (0 mg L^{-1} coating), C1 (50 mg L^{-1} coating), S0 (0 mg L^{-1} seed soaking), S1 (50 mg L^{-1} seed soaking). Different lowercase letters above bars indicate significant differences among treatments within the same variety (LSD test, $p < 0.05$)



activity of both cultivars under the moderate level; the C1 treatment elevated the average CAT activity of both cultivars, and the C0 treatment also increased the average CAT activity of Yuxiangyouzhan. The S1 treatment raised the MDA content of Xiangyaxiangzhan under moderate and low levels, and increased the MDA content of Yuxiangyouzhan under the low level (Fig. 9).

Photosynthesis pigments

V, T, and their $V \times E$, $E \times T$ and $V \times E \times T$ interactions had an extremely significant effect on the photosynthetic pigment contents (chlorophyll a, chlorophyll b, carotenoids and total chlorophyll) at 12 d; except for E and $V \times T$ (Table 1). Yuxiangyouzhan exhibited lower levels of mean photosynthetic pigments compared to Xiangyaxiangzhan. Interestingly, the application of straw extract led to an increase in mean photosynthetic pigment content in Xiangyaxiangzhan. Furthermore, the S1 (seed soaking) treatment significantly increased the mean content of Chlorophyll a, Chlorophyll b, Carotenoids, and Total chlorophyll in both cultivars at 12 d. Specifically, for Xiangyaxiangzhan, photosynthetic pigment contents were increased by the S1 treatment under the low level, by the C0 treatment under the moderate level, and by the S0 and S1 treatments under the high level. For Yuxiangyouzhan, photosynthetic pigment contents were improved by the S0 and S1 treatments under moderate and high levels (Fig. 10).

Correlation analysis

The analysis revealed that the germination rate after sowing was significantly and positively correlated ($p < 0.01$) with the biomass and plant growth at 7 d and 12 d. A significant positive correlation ($p < 0.01$) was observed among the biomass parameters of different plant tissues at 12 d. The root dry weight at 12 d showed a highly significant positive correlation ($p < 0.01$) with both the total plant dry weight at 12 d and the root to shoot ratio at 12 d. Plant height at 12 d was negatively correlated with other plant growth parameters ($p < 0.05$). The MDA content at 7 d was significantly negatively correlated ($p < 0.05$) with both the germination rate and root dry weight at 7 d. The POD activity at 12 d was also negatively correlated with shoot dry weight and dry weight per plant height at 12 d. The photosynthesis pigments was positively correlation with pod activity at 12 d (Fig. 11).

Discussion

In recent years, straw return has become a pivotal agronomic practice for sustainable agriculture, as it helps improve the physical and chemical properties of soil as well as soil fertility quality (Bi et al. 2010; Mo et al. 2025). However, its application is often constrained by allelopathic effects that

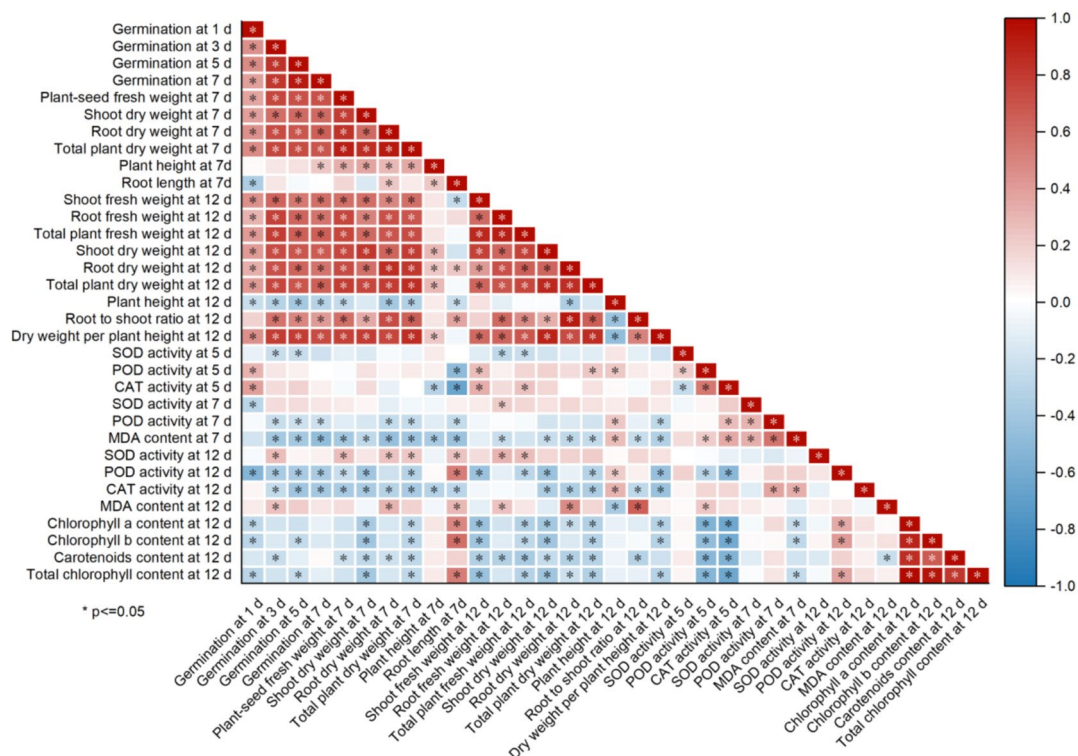


Fig. 11 Correlation analysis among the investigated parameters. An asterisk (*) indicates a statistically significant correlation at $p \leq 0.05$

can inhibit crop establishment (Lin et al. 2024; Li et al. 2025a, b). Zinc is an essential nutrient for various plant growth processes, including enhancing chlorophyll levels, improving enzyme function, and serving as a structural component of many proteins (Al Jabri et al. 2022). Nano-ZnO oxide seed coating can promote plant yield (Raha and Ahmaruzzaman 2022). Our study elucidates the complex interactions between rice straw allelochemicals, seed treatments such as nano-ZnO seed treatments, and varietal responses during the critical germination and early seedling stages.

Previous studies have shown that high-concentration rice straw extract significantly reduces plant germination (Afridi et al. 2014), and wheat straw extract can inhibit the germination of Bermudagrass by 61.7% (Alghamdi et al. 2022). The results of this experiment are consistent with previous findings (Li et al. 2024; Afridi et al. 2014; Alghamdi et al. 2022; Khaliq et al. 2012). Our results confirm the general inhibition of early germination in rice seedlings (Fig. 1), highlighting the vulnerability of this initial stage. Notably, the two cultivars exhibited divergent strategies: Yuxiangyouzhan maintained a higher baseline germination rate, whereas Xiangyaxiangzhan demonstrated greater plasticity and responsiveness to seed treatments. This genotypic specificity underscores that the allelopathic impact is modulated by inherent physiological traits. It has been reported that the application of nano-ZnO to seeds can improve germination rate, promote root development, and enhance stress resistance (Nayab et al. 2025). Studies have also found that ZnO synthesized by green and chemical methods can promote seed germination of crops such as rice, wheat, maize, sorghum and chickpea (Donia and Carbone 2023; Singh et al. 2023). The results of this study are consistent with previous research, showing that zinc oxide nanoparticles can initially improve seed germination, and our study reveals a more nuanced scenario. Nano-ZnO, particularly via seed soaking (S1), exerted an effective mitigating effect on the sensitive Xiangyaxiangzhan, significantly increasing its early germination rate even under straw extract stress, with the best seed treatment effect observed under the moderate straw extract level (Fig. 1A). This suggests that nano-ZnO may alter seed coat permeability or prime early metabolic pathways, counteracting the phytotoxic blockade. Conversely, for Yuxiangyouzhan, benefits were more conditional, evident primarily under high straw stress (H level) with specific treatments (C0, C1, S0) at 1 d (Fig. 1A), indicating that nano-ZnO oxide coating has a certain promoting effect on rice under straw extract stress, but the role of varietal factors must be considered, with a genotype-dependent bioavailability or tolerance threshold to combined treatments.

Previous studies have reported that high-concentration rice straw extract significantly inhibits plant growth (Afridi et al. 2014). In this experiment, the initial growth advantage

of Yuxiangyouzhan was significantly eroded by straw extract at 7 d, confirming the strong inhibitory effect of allelopathy on early biomass accumulation. Our results align with the classic "low-promotion, high-inhibition" model of allelopathy (Hu et al. 2012). Although nano-ZnO coating (C1) inhibited the shoot dry weight of Xiangyaxiangzhan at 7 d and 12 d after sowing and reduced the overall shoot biomass in this experiment, this treatment strategically increased the root dry weight and total dry weight of Xiangyaxiangzhan at 7 d and 12 d; however, its effect on Yuxiangyouzhan was related to growth time, mainly inhibiting dry matter accumulation in shoots at the early stage and exerting the main effect on roots at the later stage (Figs. 2, 3). This reallocation is critical, as our correlation analysis identified root dry weight as the core driver for later total biomass and a high root to shoot ratio (Fig. 6). By 12 d, as the acute stress potentially lessened or seedlings acclimated, Yuxiangyouzhan regained its growth superiority. However, the legacy of effective seed treatments was evident in Xiangyaxiangzhan, where all treatments significantly elevated the root to shoot ratio by 4.19–31.05%, promoting a more robust and resource-efficient architecture (Fig. 6). This shift towards root investment, corroborated by the negative correlation between plant height and root to shoot ratio (Fig. 6), represents a key adaptive trade-off to prioritize soil exploration and stability over rapid vertical expansion.

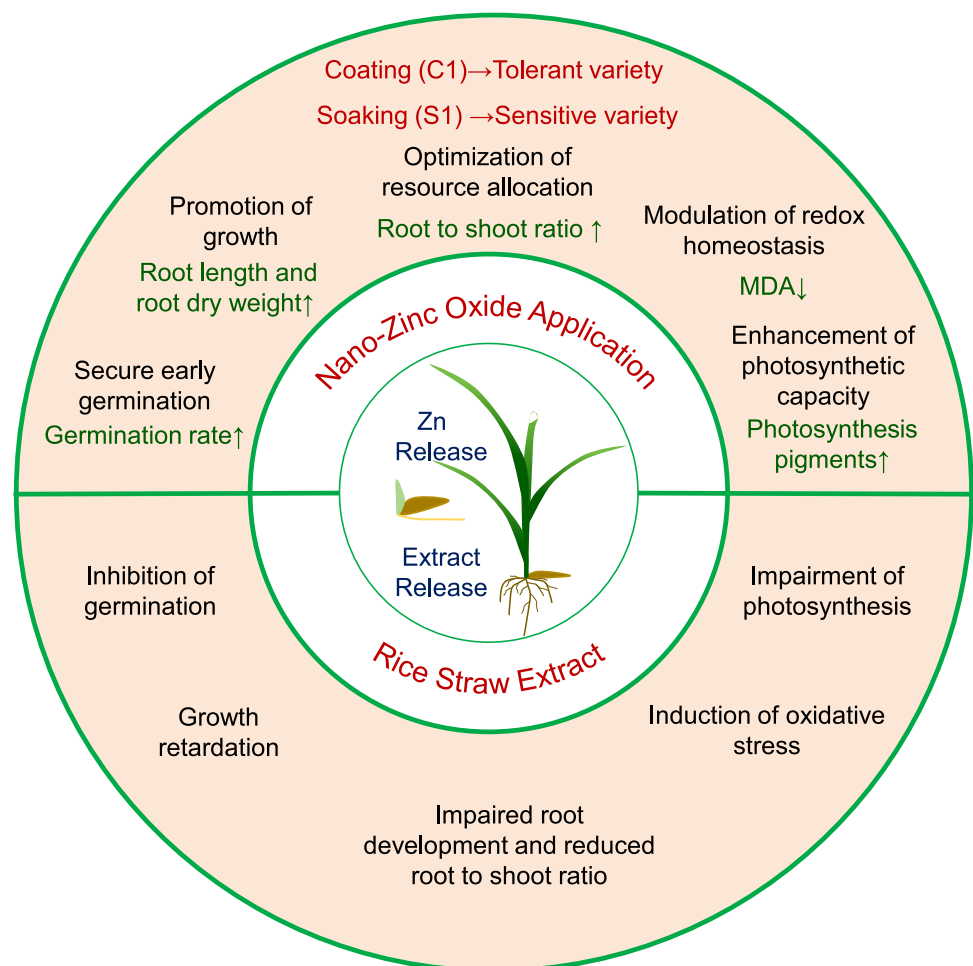
Studies have found that wheat straw extract can enhance the antioxidant enzyme activity of wheat (Khaliq et al. 2012). The higher the extract concentration, the higher the CAT and POD values, while SOD activity decreases at 100% extract concentration. It has also been shown that wheat straw extract can inhibit the growth of Bermudagrass and reduce the contents of chlorophyll a and chlorophyll b as well as SOD activity in leaves (Alghamdi et al. 2022). The antioxidant system displayed a dynamic temporal and varietal response. The general reduction in the activity SOD, POD, and the content of MDA at 7 d under straw extract might indicate an initial suppression of metabolism or a successful attenuation of oxidative burst. The differential responses of the two cultivars at 12 d, Xiangyaxiangzhan showed promotion at low concentrations and inhibition at high concentrations, while Yuxiangyouzhan exhibited an inhibitory effect, highlighted fundamental differences in their long-term oxidative stress management (Li et al. 2014; Chen et al. 2016). Nano-ZnO treatments showed a dual capacity: they could induce oxidative stress (as seen in reduced SOD under L-C1), but also, in optimized combinations like M-C1, significantly alleviate the straw-induced oxidative load in Yuxiangyouzhan (reduced POD and MDA). This aligns with the model that nanoparticle toxicity is rooted in ROS generation (Kong et al. 2019), but at optimal doses/timing, they may enhance the plant's scavenging capacity. The most integrative finding lies in the photosynthetic apparatus. Multiple

studies have shown that the use of nano-ZnO is beneficial to various plants in increasing photosynthetic pigment content, fruit quality and activating defense systems (Aigbe and Osibote 2024; Lourenço et al. 2023). Meanwhile, the allelopathic effect of straw return on crops is closely related to nutrient content, and high nutrient content can reduce the inhibition of allelochemicals on crop growth (Cheynier et al. 2013). Our study found that the S1 treatment universally increased the contents of chlorophyll and carotenoids in both cultivars, even under high straw stress (Fig. 10). This is pivotal because our correlation analysis directly links higher total chlorophyll content to greater total biomass and lower MDA content (Figs. 8, 9). This creates a positive feedback loop: nano-ZnO-augmented photosynthetic capacity not only fuels growth but also bolsters the antioxidant system, thereby "coupling" productivity with stress tolerance. The positive correlation between photosynthetic pigments and POD activity further suggests a coordinated upregulation of protective mechanisms.

In conclusion, our study demonstrates that the early growth compromise imposed by straw allelopathy is not inevitable. It can be effectively reshaped through the

synergistic use of nano-ZnO seed treatments and varietal selection. The treatment efficacy is highly context-dependent, governed by the triad of variety, nano-particle application method (coating vs. soaking), and stress intensity. For sensitive varieties like Xiangyaxiangzhan, nano-ZnO seed soaking (S1) emerges as a powerful tool to secure early stand establishment and steer resource allocation toward a resilient root system. For inherently more tolerant varieties like Yuxiangyouzhan, targeted nano-ZnO coatings (C1) under moderate stress can alleviate specific physiological damages. The core mechanism appears to be the dual role of nano-ZnO in moderating oxidative stress and supporting photosynthetic integrity, thereby uncoupling growth from stress-induced damage. These insights provide a strong theoretical foundation for designing precision seed-priming strategies to harness the benefits of straw return while mitigating its early-stage penalties, contributing to more sustainable and resilient rice production systems (Fig. 12).

Fig. 12 Mechanism of nano-ZnO oxide application and rice straw extract effects on rice germination, seedling growth, and antioxidant attributes



Conclusion

His study demonstrates that different rice cultivars exhibit distinct sensitivities to straw extract concentrations, and nano-ZnO oxide seed treatments can alleviate the early-growth constraints imposed by rice straw residues through genotype-dependent physiological rebalancing. For the more responsive cultivar Xiangyaxiangzhan, seed soaking with 50 mg L⁻¹ nano-ZnO effectively promoted the early germination rate and increased photosynthetic pigment contents. In both cultivars, seed coating with 50 mg L⁻¹ nano-ZnO enhanced root allocation and improved oxidative stress management under straw-mediated stress. The underlying mechanism involves nano-ZnO-induced shifts in biomass partitioning toward roots and the up-regulation of key antioxidant enzymes, which together help maintain photosynthetic integrity and support seedling growth. These results offer a practical nano-priming strategy to improve seedling vigor and early establishment in rice systems incorporating straw return. Taking seed treatment with 50 mg L⁻¹ nano-ZnO as an example, it can alleviate the stress caused by straw return and promote the growth of rice seedlings.

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Data availability The original contributions presented in the study are included in the article/supplementary material. Further inquiries can be directed to the corresponding author.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Institutional review board statement Not applicable.

References

- Afridi RA, Khan MA, Gul H, Khan MD (2014) Allelopathic influence of rice extracts on phenology of various crops and weeds. *Pak J Bot* 46(4):1211–1215
- Aigbe UO, Osibote AO (2024) Green synthesis of metal oxide nanoparticles, and their various applications. *J Hazard Mater Adv* 13:100401–104450
- Al Jabri H, Saleem MH, Rizwan M, Hussain I, Usman K, Alsafran M (2022) Zinc oxide nanoparticles and their biosynthesis: overview. *Life Basel* 12:594
- Alghamdi SA, Al-Nehmi AA, Ibrahim OHM (2022) Potential allelopathic effect of wheat straw aqueous extract on bermudagrass noxious weed. *Sustainability (Basel)*. <https://doi.org/10.3390/su142315989>
- Bi YY, Wang YJ, Gao CY (2010) Quantity and regional distribution of main crop straw resources in China. *J Agric Mech Res* 32(3):1–7
- Chen C, Zhao Y, Chen Z, Tang R, Qiu C (2024) Research progress on effects of nano-ZnO oxide on plant growth and development. *Jiangsu Nongye Kexue* 52(19):25–33
- Chen Y, Wang HY, Lyu XF, Ma HJ, Li QZ, Cui JH (2016) Effects of maize straw extract on seed germination and seedling physiological metabolism of three crops. *J Maize Sci* 24(4):98–104
- Cheyrier VER, Comte G, Davies KM, Lattanzio V, Martens S (2013) Plant phenolics: recent advances on their biosynthesis, genetics, and ecophysiology. *Plant Physiol Biochem* 72:1–20
- Donia DT, Carbone M (2023) Seed priming with zinc oxide nanoparticles to enhance crop tolerance to environmental stresses. *Int J Mol Sci* 24(24):17612
- Du K, Xie Y, Lin Z, Liu Z, Wang S, Ding Y (2016) Effects of nitrogen fertilizer on rice seedling growth and nutrient uptake under straw returning conditions. *J Nanjing Agric Univ* 39(01):18–25
- Elhaj BZ, Unrine JM (2018) Functionalized-ZnO-nanoparticle seed treatments to enhance growth and Zn content of wheat (*Triticum aestivum*) seedlings. *J Agric Food Chem* 66(46):12166–12178
- Elshayb OM, Nada AM, Ibrahim HM, Amin HE, Atta AM (2021) Application of silica nanoparticles for improving growth, yield, and enzymatic antioxidant for the hybrid rice EHR1 growing under water regime conditions. *Materials* 14:1150
- Hu SK, Gao Y, Wang X (2012) Study on autotoxicity of rice stubble. *Mod Agric Sci Technol* 571(5):51–52
- Huang Z, Xie W, Wang M, Liu X, Ashraf U, Qin D, Zhuang M, Li W, Wang S, Tian H, Mo Z (2020) Response of rice genotypes with differential nitrate reductase-dependent NO synthesis to melatonin under ZnO nanoparticles' (NPs) stress. *Chemosphere* 250:126337
- Khalik A, Matloob A, Aslam F, Mushtaq MN, Khan MB (2012) Toxic action of aqueous wheat straw extract on horse purslane. *Planta Daninha* 30(2):269–278
- Kong ZH, Wang ZY, Chen N, Zhou AR, Zeng SX, Lin SL (2019) Research progress on application and toxicology of nano-ZnO in food. *Storage Process* 19(2):174–178
- Li B, Wu WW, Shen WY, Xiong F, Wang KH (2024) Allelochemicals released from rice straw inhibit wheat seed germination and seedling growth. *Agronomy* 14:2376
- Li F, Ding H, Meng A, Bi G, Lan W, Guo Q, Zhang G, Qin F, Zhang Z, Zhang J (2025a) Effects of nano-ZnO oxide seed soaking on germination and physiological characteristics at flowering and pegging stage of peanut under drought stress. *J Peanut Sci* 54(03):266–272
- Li G, Zhang HY, Wang XL (2014) Effects of rice straw extract on wheat and associated weeds growth. *Weed Sci* 32(1):34–38
- Li J, Gao S, Bao C, Yan S, Ma C, Ma C, Yan C (2025b) Mechanisms by which soil solution Fe²⁺ affects seedling growth of rice under rice straw return. *Agronomy* 15(2):271

- Li YZ, Liang LX, Li W, Ashraf U, Ma L, Tang XR, Pan SG, Tian H, Mo ZW (2021) ZnO nanoparticle-based seed priming modulates early growth and enhances physio-biochemical and metabolic profiles of fragrant rice against cadmium toxicity. *J Nanobiotechnol* 19(1):75
- Lin T, Chen X, Ren Y, Qing B, Zhang M, Mo Z, Wang S (2024) Effect of iron oxide nanocoating on the seed germination, seedling growth, and antioxidant response of aromatic rice grown in the presence of different concentrations of rice straw extracts. *J Nanopart Res* 26:78
- Liu D, Song C, Xin Z, Fang C, Liu Z, Xu Y (2023) Agricultural management strategies for balancing yield increase, carbon sequestration, and emission reduction after straw return for three major grain crops in China: a meta-analysis. *J Environ Manage* 340:117965
- Liu RQ, Zhang HY, Lal R (2016) Effects of stabilized nanoparticles of copper, zinc, manganese, and iron oxides in low concentrations on lettuce (*Lactuca sativa*) seed germination: nanotoxicants or nanonutrients? *Water Air Soil Pollut* 227(1)
- Lourengo IM, Freire BM, Pieretti JC, Reis RAD, Soares NM, Santos MDL, Batista BL, Seabra AB, Lange CN (2023) Implications of ZnO nanoparticles and S-nitrosoglutathione on nitric oxide, reactive oxidative species, photosynthetic pigments, and ionic profile in rice. *Antioxidants* 12:1871
- Lucca P, Poletti S, Sautter C (2006) Genetic engineering approaches to enrich rice with iron and vitamin A. *Physiol Plant* 126:291–303
- Ma Y, Shen Y, Liu Y (2020) State of the art of straw treatment technology: challenges and solutions forward. *Bioresour Technol* 313:123656
- Mahakham W, Sarmah AK, Maensiri S, Theerakulpisut P (2017) Nanopriming technology for enhancing germination and starch metabolism of aged rice seeds using phytosynthesized silver nanoparticles. *Sci Rep* 7:1–21
- Mo ZW, Cheng SR, Ren Y, He LX, Pan SG, Liu HD, Tian H, Ashraf U, Duan MY, Tang XR (2025) Reduced tillage coupled with straw return improves the grain yield and 2-acetyl-1-pyrroline content in fragrant rice. *J Integr Agric* 24(5):1718–1737
- Nayab A, Sidra A, Afife BUK, Sezai E, Mian AA, Adewale AS, Ghulam M, Humaira R, Saeed MA, Rashid I (2025) Enhancement of rice zinc content using green synthesized ZnO-NPs by foliar and nano-priming applications. *Appl Biochem Biotechnol* 197:1906–1922
- Pedrin S, Merritt DJ, Stevens J, Dixon K (2017) Seed coating: science or marketing spin? *Trends Plant Sci* 22(2):106–116
- Peng J, Wang DY, Xu CM, Chen LP, Deng F, Zhang XF (2009) Determination of catalase activity in rice by ammonium molybdate method. *Chin Agric Sci Bull* 25(16):61–64
- Qing BW, Jiang Y, Chen YJ, Chen JL, Xie HJ, Mo ZW (2022) Nitrogen modulates early growth and physio-biochemical attributes in fragrant rice grown under cadmium and multiwall carbon nanotubes stresses. *Environ Sci Pollut Res* 29(45):67837–67855
- Raha S, Ahmaruzzaman M (2022) ZnO nanostructured materials and their potential applications: progress, challenges and perspectives. *Nanoscale Adv* 4:1868–1925
- Singh A, Sengar RS, Rajput VD, Agrawal S, Ghazaryan K, Minkina T (2023) Impact of zinc oxide nanoparticles on seed germination characteristics in rice (*Oryza sativa* L.) under salinity stress. *J Ecol Eng* 24(10):142–156
- Sun Y, Hu W (2025) Effects of nano-ZnO oxide seed soaking on seedling quality of rice, wheat and maize. *Agric Technol Equip* 02:171–172+177
- Tonday M, Kalia A, Singh A, Dheri GS, Taggar MS, Nepovimova E, Krejcar O, Kuca K (2021) Seed priming and coating by nano-scale zinc oxide particles improved vegetative growth, yield and quality of fodder maize (*Zea mays*). *Agronomy* 11(4):729
- Venkidasamy B, Safia N, Preethy SK, Thiruvengadam M, Sathishkumar R (2018) Assessment of the effects of metal oxide nanoparticles on the growth, physiology and metabolic responses in vitro grown eggplant (*Solanum melongena*). *3 Biotech* 8(8):362
- Wang S, Liu YH, Ren L, Zhang Z, Li F, Gu DY, Kong FL (2021) Growth and physiological characteristics of *Eichhornia crassipes* and *Acorus calamus* under stress of nano-ZnO with different particle sizes. *Wetl Sci* 19(3):317–324
- Xu X, Dong Y, Fu W (2024) Effects of phenolic allelochemicals on nutrient supply capacity in rhizosphere soil. *J Ecol Rural Environ* 40(10):1358–1365
- Xue Q, Han B, Wu X, Li P, Li Y, Wu Q (2020) Application and prospect of nanomaterials in crop production. *Biotechnol Bull* 10(06):655–660
- Yang L, Zhang L, Zhang Q, Wei J, Zhao X, Zheng Z, Chen B, Xu Z (2024) Nanopriming boost seed vigor: deeper insights into the effect mechanism. *Plant Physiol Biochem* 214:108895
- Yin X, Zhu J, Gu D, Yang W, Du X, Wen T, Sun A, Qian X, Shi Y (2021) Effects of nano-ZnO oxide on rice seed germination and seedling growth. *Jiangsu Agric Sci* 49(19):101–106
- Yu JG, Gu Y, Chang ZZ (2013) Allelopathic effects of wheat straw extract and decomposed solution on rice. *Soil Sci J* 50(02):349–356
- Zhang RP, Yu HY, Zhang WB, Li W, Su H, Wu SX, Xu Q, Li YY, Yao HY (2024) Straw return enhances grain yield and quality of three main crops: evidence from a meta-analysis. *Front Plant Sci* 15:1433220
- Zhao XQ, Chen Y, Li HS, Lu JY (2021) Influence of seed coating with copper, iron and zinc nanoparticles on growth and yield of tomato. *IET Nanobiotechnol* 15(8):674–679

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